CSC 1300 LAB 6

SPRING 2026

Concepts

- Global constants
- Functions, sending data to a function

Lab Policies - What you Can't Do/Use in this lab

If you do any of the following, you will receive a 0 on some or all of the assignment and possibly get charged with an academic integrity violation (this is very serious).

1. You may not use code or syntax in your lab assignment **that has not been taught** in CSC 1300 class **without referencing your source**. How do you reference your source? By providing a link/URL in a multi-line comment in the section of code where you use "non-1300" code. Otherwise, this will be considered plagiarism. It is safer to use course materials (slides & example programs) to help you create solutions to the assignments.

2. You may not (ever) use the following in your programming & lab assignments in CSC 1300:
   - while(true) or while(1)
   - goto statements
   - recursion
   - exit()
   - ternary operators
   - Any syntax requiring a '?'
   - lambda functions
   - try-catch

3. You **may not use generative AI (genAI)** **<u>in any way</u>** for this lab assignment. Not sure if something is GenAI? Ask a TA or instructor!

4. You may not share your code with anyone (except lab instructors/TAs/instructors).

Lab Report

<u>After</u> you have completed this assignment, fill out the lab report and submit your lab report in your zip file along with your source files. Name your screenshot **lab6ReportProof**.
Lab Report Link: **https://tntech.co1.qualtrics.com/jfe/form/SV_6M4wwN1g5Ynv01w**

What to Submit in ilearn when you are done

Create a **zip** file named **username_lab6** containing the following .cpp files and upload it to ilearn in the **Lab 6** assignment.

- `Lab6.cpp`
- `Lab6ReportProof`

Specifications

- Create a menu-based program that will allow the user to calculate the area for a few different shapes: square, rectangle, parallelogram, and circle.

- Refer to the Sample Output in this document to see what menu options you should use.

- Create ***PI*** as **a global constant** variable.

- **You should not have any other variables be defined globally. Major points will be deducted for non-constant global variables.**

- Main Function
  - use do-while to repeat program until user chooses option 5
  - call the displayMenu function (and accept the user's choice that was returned)
  - depending on user's choice, get the data required to calculate the area of the shape from the user and then call the appropriate function to calculate the area.

- Write five programmer-defined functions.
  - **displayMenu()** – should print out the menu options, get the user's choice, & validate with while loop. Then return user's choice from this function.
  - **areaSquare**() – should accept the height of the square from the main function, calculate the area, and then print the area of the square.
  - **areaRectangle**() – should accept the height & width of the rectangle from the main function, calculate the area, and then print the area of the rectangle.
  - **areaParallelogram**() – should accept the base & height of the parallelogram from the main function, calculate the area, and then print the area of the parallelogram.
  - **areaCircle**() – should accept the radius of the circle from the main function, calculate the area, and then print the area of the circle.

- Save your program as **lab6.cpp**.

Sample Output

```
Choose a shape to calculate area or 5 to quit:
        1. Square
        2. Rectangle
        3. Parallelogram
        4. Circle
```

```
        5. End Program
CHOOSE 1-5: 1

Enter the height of the square:
3.5

The area of your square is 12.25


Choose a shape to calculate area or 5 to quit:
        1. Square
        2. Rectangle
        3. Parallelogram
        4. Circle
        5. End Program
CHOOSE 1-5: 2

Enter the height & width of the rectangle, separated by a space:
9.3
3.2

The area of your rectangle is 29.76


Choose a shape to calculate area or 5 to quit:
        1. Square
        2. Rectangle
        3. Parallelogram
        4. Circle
        5. End Program
CHOOSE 1-5: 3

Enter the base and height of the parallelogram, separated by a space:
5.6
3.2

The area of your parallelogram is 17.92


Choose a shape to calculate area or 5 to quit:
        1. Square
        2. Rectangle
        3. Parallelogram
        4. Circle
        5. End Program
CHOOSE 1-5: 4

Enter the radius of the circle:
94.5

The area of your circle is 28055.2


Choose a shape to calculate area or 5 to quit:
        1. Square
        2. Rectangle
        3. Parallelogram
        4. Circle
        5. End Program
CHOOSE 1-5: -4

Invalid input.
CHOOSE 1-5: 5
```

Validate user input!

```
Choose a shape to calculate area or 5 to quit:
        1. Square
        2. Rectangle
        3. Parallelogram
        4. Circle
        5. End Program
CHOOSE 1-5: dog

Invalid input.
CHOOSE 1-5: 5


Goodbye!
```

Validate user input!